

Forward Guidance and the Dynamics of Bank Credit

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FORWARD GUIDANCE AND BANK CREDIT

Motivation: During the 2008–2015 zero bound, forward guidance (FG) became an important policy tool for the Fed

Question: Does FG stimulate bank credit, and how so?

Finding: *Asymmetry* — FG effectively **contracts** credit, but does not **expand** it much
⇒ FG is least effective at stimulating credit precisely when most needed

Transmission: news about the future policy path moves asset prices immediately

- ▶ Surprise **tightening**: prices ↓, yields ↑, leverage ↑, lending ↓
- ▶ Surprise **easing**: signs flip — but statistically indistinguishable from zero

Data

High-frequency identification

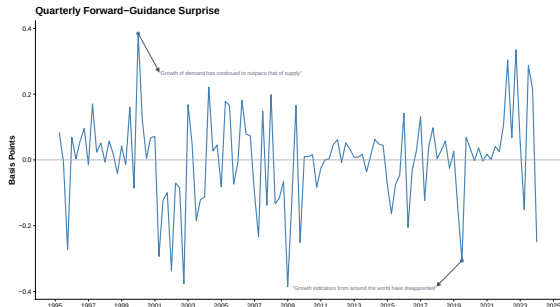
- ▶ 30-min futures window around FOMC announcements (Gürkaynak et al. 2005; Nakamura & Steinsson 2018)

Bank Data

- ▶ 1995 Q3 - 2019 Q4, quarterly
- ▶ Sources: FFIEC Call Reports, FDIC, FRB

Controls

- ▶ Macro: FFR, RGDP, RINV, BAA-10Y, GDP Defl.
- ▶ Bank: Tier 1 ratio, RWA/Assets, Deposits/Assets, AOCI/Equity



▶ FG Shock Series Interpretation:

- ▶ Positive spike: contractionary signal
- ▶ Negative spike: expansionary signal

PARSING OUT THE SHOCK

Signals about the *future* policy-rate path move bank balance sheets *today*, but the two directions are not mirror images.

Contractionary signal ($FG_t > 0$)

- ▶ Higher expected future rates $\Rightarrow$ tighter funding
- ▶ Equity falls faster than assets $\Rightarrow$ leverage $\uparrow$, lending $\downarrow$
- ▶ *Immediate, front-loaded response*

Expansionary signal ($FG_t < 0$)

- ▶ Lower expected future rates $\Rightarrow$ cheaper funding
- ▶ Balance sheets barely move on impact
- ▶ *Weak, back-loaded response*

BASLINE EMPIRICAL SPECIFICATION: LOCAL PROJECTIONS

For each horizon $h = 0, 1, \dots, 20$ estimate:

$$\Delta_{\tau} \log(y_{i,t+h}) = \alpha_{h,i} + \beta_h^C FG_t^C + \beta_h^E FG_t^E + \sum_{j=1}^4 \Gamma_{h,j} Z_{t-j} + \varepsilon_{i,t+h}$$

Where $\Delta_h \log(y_{i,t+h}) = \log(y_{i,t+h}) - \log(y_{i,t-1})$

Outcome Variable: $y_{i,t}$ is either loans or leverage

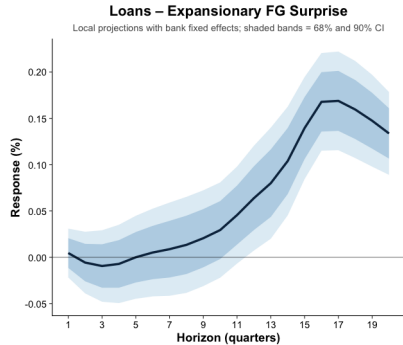
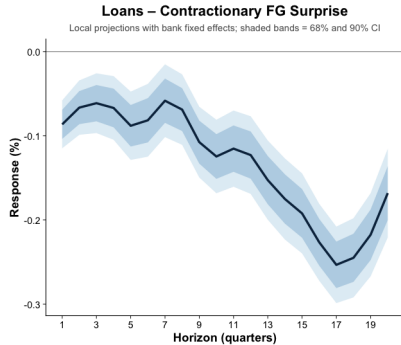
Bank-level fixed effects: $\alpha_{h,i}$

Controls: $Z_{i,t-j}$

FG Shocks: Indicators FG_t^C for contractionary FG shock and FG_t^E for expansionary FG shock at time t

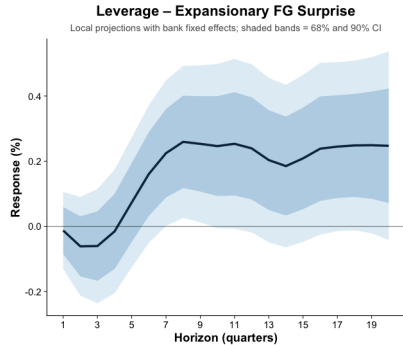
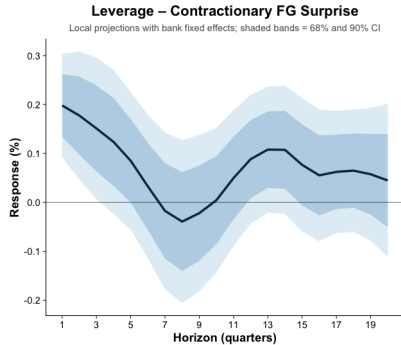
Inference: Standard errors are Driscoll-Kraay

LENDING IRFs



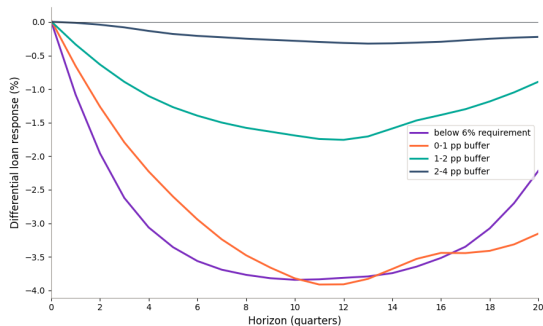
The figures are the impulse responses of total loans for banks to a one standard deviation contractionary (left) and expansionary (right) forward guidance shock. The inner dark band corresponds to the 68% confidence band, while the outer light band corresponds to the 90% confidence band.

LEVERAGE IRFs



The figures are the impulse responses of leverage for banks to a one standard deviation contractionary (left) and expansionary (right) forward guidance shock. The inner dark band corresponds to the 68% confidence band, while the outer light band corresponds to the 90% confidence band.

CAPITALIZATION CHANNEL IN THE LENDING RESPONSE



Differential response to a contractionary FG shock; reference group is banks
with Tier 1 buffer $\geq 4\%$

Question. Does the loan response depend on proximity to regulatory capital constraints?

Empirical design

- ▶ Sort banks by lagged Tier 1 buffer relative to the 6% requirement
- ▶ Interact contractionary FG shocks with buffer-level indicators

Pattern in the IRFs

- ▶ Credit contraction is larger when capital is tighter.

Takeaway Contractionary FG works through a capitalization channel: tight capital positions amplify credit-supply cuts

MODEL: FINANCIAL FRICTIONS

Banks face

- ▶ Budget Constraint: $L_t + B_t + R_t = D_t + (1 - f(\delta_t))X_t$ where $\delta_t \equiv \frac{L_t}{X_t}$
- ▶ Cost of Raising Equity: $f(\delta_t) = \frac{\alpha}{2}\delta_t^2$
- ▶ Reserve Requirement: $R_t \geq \rho D_t$ $0 \leq \rho < 1$
- ▶ Capital requirement: $\frac{L_t}{X_t} \equiv \delta_t \leq \bar{\delta}$

Firms face

- ▶ Marginal Cost: $(1 + i_t^L)[1 - \phi_{d,t+1}(j)]W_t(j)$
- ▶ Possibility of Default: $\phi_{d,t+1}(j) = \max\left(1 - \frac{Y_t(j)}{(1 + i_t^L)}, 0\right)$

SIGNALS AND BELIEF FILTERING

Forward guidance as a set of policy signals
(Campbell et al. 2019)

Central bank's policy rate:

$$i_t^R = \phi_\pi \pi_t + \phi_y y_t + \theta_t,$$

$$\theta_t = \sum_{j=1}^{20} \varepsilon_{t-j}^{(j)}$$

- ▶ θ_t : represents past deviations from the policy rule
- ▶ $\varepsilon_t^{(j)}$: j -period **policy rule deviation** — announced at t but realized in $t + j$

Forward Guidance Shock:

$$\epsilon_t^{FG} = [s_t^0 \ s_t^1 \ \dots \ s_t^{20}]$$

A vector of noisy **signals** about future policy deviations

Noisy observation of each future component:

$$s_t^j = \varepsilon_t^j + \eta_t, \quad \eta_t \sim N(0, \sigma^2)$$

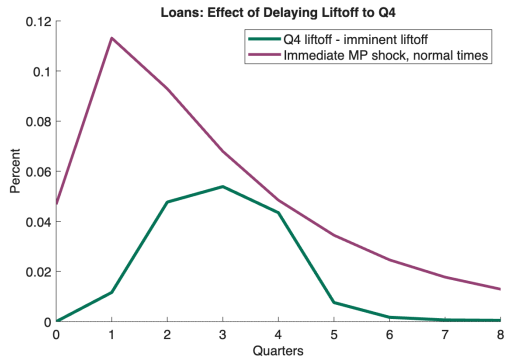
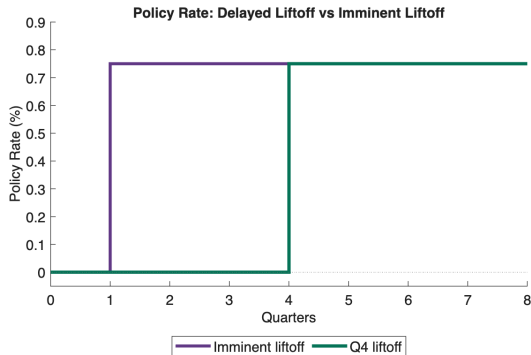
With perfect information about today's deviation:

$$s_t^0 = \epsilon_t^0$$

Belief updating (via Kalman filter):

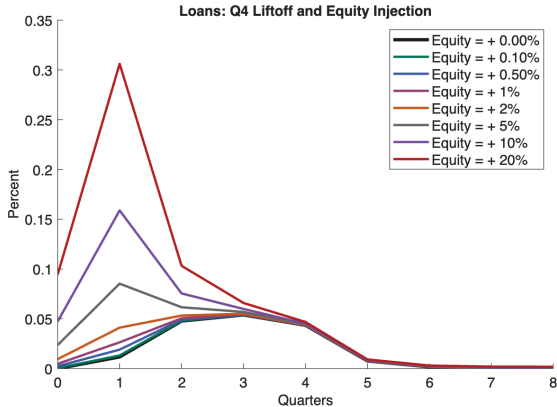
$$E_t \theta_{t+j} = E_{t-1} \theta_{t+j} + K s_t^j$$

ZERO BOUND



Left: policy rate is announced to remain at zero for three more quarters. **Right:** Compares response of lending to non-ZLB MP shock (Red) to FG shock of same magnitude at ZLB (Green) resulting from policy rate expansion in left plot

BALANCE SHEET RELIEF



Response of lending to a simultaneous FG expansion (as before) and equity injection from the government, varying by percentage of steady state bank equity.

Question: What if central bank/treasury provide balance sheet relief via equity injections?

Setup

- ▶ Same FG expansion as before (3 additional quarters of ZLB)
- ▶ Government injects equity to bank's balance sheet simultaneously

Outcome

- ▶ Stimulus of expansionary FG only approaches that of ordinary MP when equity injection is about 10% of steady state bank equity
- ▶ For US as of 2026, that is roughly the size of TARP

Conclusion

CONCLUSION

Question: can forward guidance move bank credit — and is the effect symmetric?

Findings

- ▶ FG effectively **contracts** credit but cannot **expand** it — the same asymmetry holds for bank leverage
- ▶ Bank capitalization is a driver of the credit response
- ▶ A structural model with occasionally-binding constraints reproduces these patterns

Forward guidance is least effective at stimulating credit precisely when it is needed most

Thank You

Appendix

CONSTRUCTION OF MP SHOCK

Eurodollar Futures

- ▶ Eurodollars: are U.S. dollars held in deposit accounts in banks outside the United States.
- ▶ Interest of Eurodollar is the rate at which banks lend Eurodollars to one another
- ▶ Eurodollar futures: instrument traders used to hedge the direction of short-term interest rates.
- ▶ Price = 100 - implied interest rate
⇒ Implied rate = 100 - Price
- ▶ Bauer & Swanson calculate:

$$\Delta ED^h = ED_{20+}^h - ED_{10-}^h$$

- ▶ ED_{20+}^h : first traded h th nearest Eurodollar future contract 20 min after FOMC announcement
- ▶ ED_{10-}^h : last traded h th nearest Eurodollar future contract 10 min before FOMC announcement

Eurodollar Futures

- ▶ There are two parties involved:
 - ▶ Short position: party expects interest rates to rise (price to fall)
 - ▶ Long position: party expects interest rates to fall (price to rise)
- ▶ Loser pays winner

Federal Funds Futures

- ▶ Similar to Eurodollar futures, except Eurodollars used LIBOR
- ▶ Fed Funds Futures use Effective Federal Funds Rate

PCA

- ▶ Track changes in prices of futures contracts around FOMC announcement
- ▶ Extract two factors — second factor is rotated so as not to load on fed funds surprise

Model

MODEL

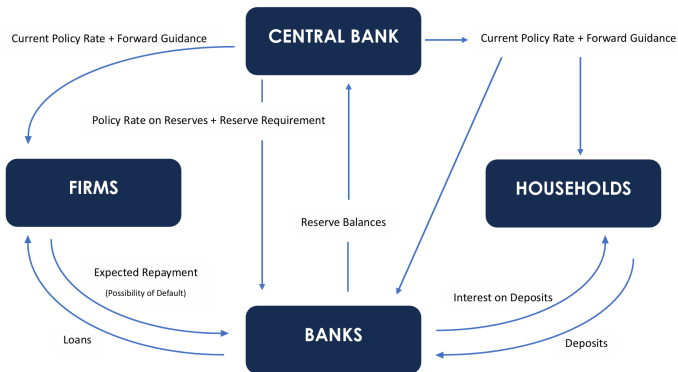


Figure 1: Model Structure

FORWARD GUIDANCE STRUCTURE

$$i_t^R = \phi_\pi \pi_t + \phi_y y_t + \theta_t,$$

$$\theta_t = \sum_{j=1}^{20} \varepsilon_{t-j}^{(j)}$$

- ▶ θ_t : represents past deviations from the policy rule
- ▶ $\varepsilon_t^{(j)}$: j -period **policy rule deviation** — announced at t but realized in $t + j$

Forward Guidance Shock:

$$\epsilon_t^{FG} = [s_t^0 \ s_t^1 \ \dots \ s_t^{20}]$$

A vector of noisy **signals** about future policy deviations

SIGNALS AND BELIEF FILTERING

Noisy observation of each future component:

$$s_t^j = \varepsilon_t^j + \eta_t, \quad \eta_t \sim N(0, \sigma^2)$$

With perfect information about today's deviation:

$$s_t^0 = \epsilon_t^0$$

Belief updating (via Kalman filter):

$$E_t \theta_{t+j} = E_{t-1} \theta_{t+j} + K_j s_t^j$$

Perfect information today:

$$E_t \theta_t = \theta_t$$

Intuition

- ▶ Agents receive a vector of signals ϵ_t^{FG} from the central bank about the future path of the policy rate i_t
- ▶ The signals contain ambiguity about the true realization of future policy rates
- ▶ These signals feed into forward looking equations that effect outcomes today, without affecting i_t today

FORWARD GUIDANCE TRANSMISSION

Say that the central bank announces a three-period ahead shock ϵ_t^3 to the policy rate

Forward Guidance Transmits Through Forward Looking Variables

Take the Euler equation:

$$\begin{aligned}c_t &= E_t[c_{t+1}] - \frac{1}{\sigma}(i_t - E_t[\pi_{t+1}]) \\&= E_t[c_{t+2} + \frac{1}{\sigma}(i_{t+1} - \pi_{t+2})] + \frac{1}{\sigma}(i_t - E_t[\pi_{t+1}]) \\&\vdots \\&= E_t\{c_{t+4} + \frac{1}{\sigma}[\phi_\pi \pi_{t+3} + \phi_y y_{t+3} + \theta_{t+3} - \pi_{t+4}]\} + \frac{1}{\sigma} \sum_{j=0}^2 (i_{t+j} - \pi_{t+j+1})\end{aligned}$$

Inserting the definition of θ_{t+3} and s_t^3 , we get:

$$c_t = E_t\{c_{t+4} + \frac{1}{\sigma}[\phi_\pi \pi_{t+3} + \phi_y y_{t+3} + \underbrace{K_3(\epsilon_t^3 + \eta_t)}_{\text{forward guidance shock}} + E_{t-1}\theta_{t+3} - \pi_{t+4}]\} + \frac{1}{\sigma} \sum_{j=0}^2 (i_{t+j} - \pi_{t+j+1})$$

FORWARD GUIDANCE WITH ASYMMETRIC KALMAN FILTERING

Belief updating (Kalman filter):

$$E_t \theta_{t+j} = E_{t-1} \theta_{t+j} + K_j s_t^j.$$

Asymmetric Kalman update (bad news hits harder)

$$K_j(s) = \begin{cases} K_j^-, & \text{if } s_t^{(j)} > 0 \text{ (contractionary news)} \\ K_j^+, & \text{if } s_t^{(j)} \leq 0 \text{ (expansionary news)} \end{cases} \quad \text{with } K_j^- > K_j^+ \geq 0,$$

Intuition: downturns tighten the capital constraints faced by banks

FINANCIAL FRICTIONS

Banks face

- ▶ Budget Constraint: $L_t + B_t + R_t = D_t + (1 - f(\delta_t))X_t$ where $\delta_t = \frac{L_t}{X_t}$
- ▶ Cost of Raising Equity: $f(\delta_t) = \frac{\alpha}{2}\delta_t^2$
- ▶ Reserve Requirement: $R_t \geq \rho D_t$ $0 \leq \rho < 1$

Firms face

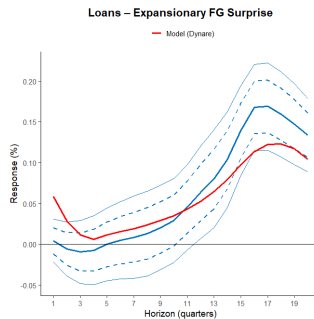
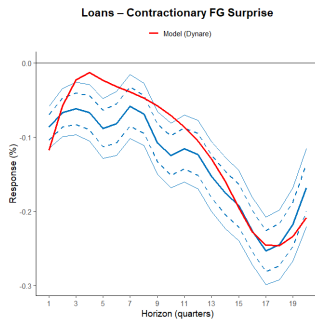
- ▶ Marginal Cost: $(1 + i_t^L)[1 - \phi_{d,t+1}(j)]W_t(j)$
- ▶ Possibility of Default: $\phi_{d,t+1}(j) = \max\left(1 - \frac{Y_t(j)}{(1 + i_t^L)}, 0\right)$

CALIBRATION

Parameter	Value	Description	Source
β	0.99	Discount Factor	–
σ	2	Risk Aversion	–
η	2	Frisch Elasticity of Labor Supply	–
θ	1	Labor Disutility Weight	–
ϕ	0.1	Liquidity Services Weight	–
ρ	0.1	Reserve Ratio	Benigno & Benigno (2021)
τ	0.3	Tax Rate	OECD Centre for Tax Policy and Administration

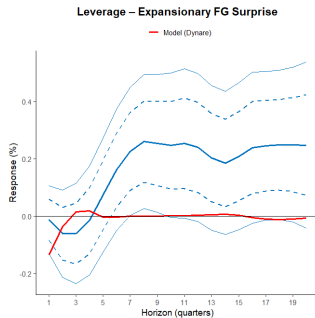
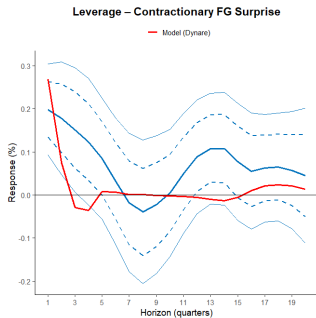
These parameters are fixed, the remainder are estimated using Bayesian methods drawing upon U.S. macro and financial data.

LOAN RESPONSES TO FG SHOCK



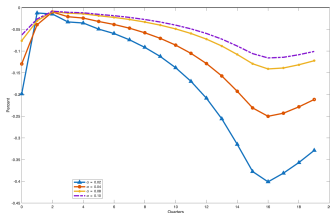
Note: The blue IRFs are data-based estimates (solid line with bands). The dashed lines represent the 68% confidence bands, and the outer solid lines represent the 90% confidence bands. The red IRF is generated by the quantitative model.

BANK LEVERAGE RESPONSES TO FG SHOCK RETURN

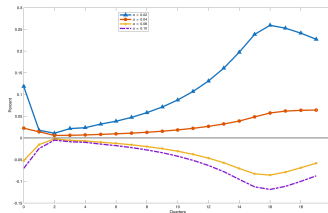


Note: The blue IRFs are data-based estimates (solid line with bands). The dashed lines represent the 68% confidence bands, and the outer solid lines represent the 90% confidence bands. The red IRF is generated by the quantitative model.

LENDING AND THE COST OF RAISING EQUITY



Lending



Leverage

Note: Varying costs of raising equity. Blue line represents most costly environment. Purple represents least costly environment.

- ▶ Deteriorating capitalization $\Rightarrow$ stronger credit contraction
- ▶ Consistent with asymmetry logic:
 - ▶ Contractionary FG pushes banks closer to capital constraint than expansionary FG
 - ▶ In model: capitalization is key driver of credit

ZERO BOUND

- ▶ To implement an expansionary FG shock at the ZLB, I force the policy rate to remain at zero for longer
- ▶ Later liftoff = more accommodative policy
- ▶ Fix the policy rate $i_t^R = 0$ by setting $\phi_\pi, \phi_y = 0$
- ▶ Introduce a shock ε_t^4 (tightening four quarters into the future)
- ▶ Separately, introduce a shock ε_t^8 (tightening eight quarters into the future)
- ▶ Take differences in impulse responses between two shocks across all horizons:

$$\Delta Y_{8,4} = Y_{t,\varepsilon_t^8} - Y_{t,\varepsilon_t^4} \equiv Y_{t,\varepsilon_t^{FG}}^{ZLB}$$

- ▶ Where $Y_{t,\varepsilon_t^{FG}}^{ZLB}$ is the impulse response of Y to an expansionary forward guidance shock at the ZLB

ZERO BOUND

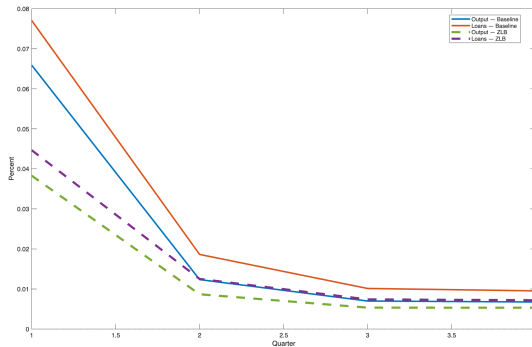


Figure 2: FG Stimulus: ZLB vs Normal Times

Note: This compares the effects of a four-quarter ahead shock in two cases: 1) ZLB FG that holds rates down for another year. 2) FG under normal times that signals a rate cut one year ahead.

POLICY RATE RESPONSE TO FG SHOCK

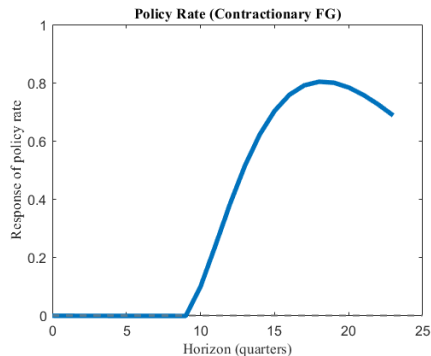
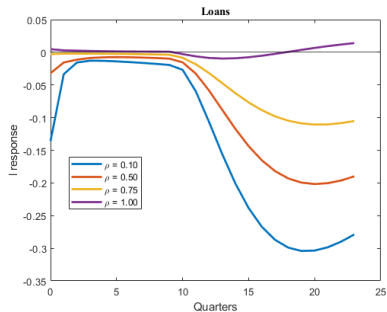


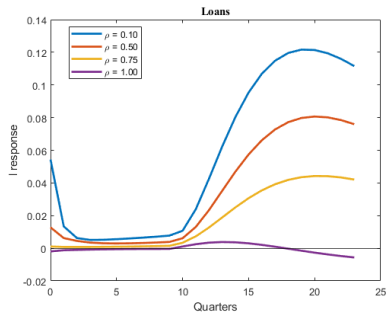
Figure 3: Policy Rate Response to Expansionary FG Shock

Note: This is the response to the policy rate after a forward guidance shock that is announced be going into effect 10 periods hence. We have no lift-off until the actual implementation of the policy rate change.

LENDING BY RESERVE RATIO



Contractionary FG Surprise



Expansionary FG Surprise

Note: ρ represents the reserve ratio. When $\rho = 0.10$, banks must hold 10% of deposits in reserves.
When $\rho = 1$, deposits are fully backed by reserves.